UNIT-1

1. Briefly explain the various components of an industrial robot. State the main function of each of the components.
2. Define automation. With the help of a figure explain the three broad classes into which industrial automation is classified.
3. How would you define a robot? Explain the two classes of industrial robots and service robots with suitable examples.
4. What are the three laws of robotics formulated by Issac Asimov?
5. The population of robot's world is increasing. What are the factors contributing to this growth?
6. Briefly explain the accuracy and repeatability specifications of robot.
7. Define Industrial Robot, coined by Robotics Institute of America and mention advantages of robots.
8. Define robot with respect to today's context. Highlight the differences between Robotics and Automation.
9. Outline the History of Robotics.
10. Illustrate Payload of a Robot.
11. Describe the Spatial Resolution of Robot and determine the increments if the robots have 8 bits of storage capacity in control memory.
12. Briefly explain the Compliance of the Robotic system.
13. Describe the Payload and Precision concepts of a robot.
14. Explain the various specifications that one should look forward to purchase a commercial robot.
15. Mention the advantages and disadvantages of robotics.
16. Distinguish between accuracy and repeatability of a robot.
17. Define speed of response.
18. Explain the relationship of robotics with industrial automation and illustrate the same with a suitable example.
19. Discuss the impact of robotic induction on direct labour.
20. List out the different specification of robot and explain any four specifications.
21. Explain Reach and Repeatability in robots.
22. List different types of robots and explain any four.
23. Explain Manipulators and Mobile robots.
24. Highlight the difference between robot and cobots.
25. Explain aerial robots and autonomous robots
26. Explain Humanoids and Swarm Robots

UNIT-2

1. You are required to design a robot that can pick up an object of arbitrary shape from a table and place it inside a box,
   a. What is the minimum number of degrees of freedom it should have?
   b. List all the possible arm configurations.
   c. Draw their corresponding work volumes.
2. Compare the size of work volumes of robots having polar, cylindrical and cartesian anatomy configuration. Assume that the robots have equal link lengths. Which Configuration has the largest work volume?
3. With a neat sketch briefly explain the Cartesian configuration of Robot.
4. With a neat labeled diagram explain the Jointed robot configuration.
5. With a neat labeled sketch briefly explain the three degrees of freedom associated with robot arm and body of a polar coordinate robot.
6. Explain the Polar configuration robot with neat sketch.
7. With a neat diagram explain the salient features of cylindrical robot configuration.
8. With a neat labeled sketch analyze the three degrees of freedom associated with robot wrist.
9. With a neat labeled diagram, describe the Cylindrical Configuration of Robot and mention type of joint movements.
10. Describe any four three work envelop of a robot with suitable diagram and mention its applications.
11. Describe any four basic robot configurations with neat sketch and narrate individual merits, demerits.
12. Describe the Degrees of freedom of robot and End effectors.
13. With a neat diagram explain the salient features of SCARA.
14. With neat sketches explain different types of joints in robot.
15. Explain with neat sketch vertical traverse, radial traverse and rotational traverse.
16. Describe with neat sketch degrees of freedom associated with a robot wrist.
17. Enumerate different Design considerations of Links & Joints.
18. Explain Forward Kinematics and Inverse Kinematics
19. What is meant by work volume?
20. With a diagram explain the anatomy of a robot.
21. The co-ordinate of a point $P_{abc}=(5, 4, 3)^T$ in the body co-ordinate frame OABC is rotated 30^0 about OZ axis. Determine the co-ordinates of the vector P_{xyz} with respect to base reference co-ordinate frame.
22. The co-ordinate of a point $P_{abc}=(7, 5, 3)^T$ which is rotated about OX axis of the reference frame OXYZ, by angle of 60^0. Determine the co-ordinates of the vector P_{xyz}.
23. The co-ordinate of a point $P_{abc}=(4, 3, 2\sqrt{3})^T$ which is rotated about OY axis of the reference frame OXYZ, by angle of 60^0. Determine the co-ordinates of the vector P_{xyz}.
24. The co-ordinate of a point $P_{abc}=(2, 3, 4)^T$ has to be translated through distance of 4 units along OX axis and -2 units along OZ axis. Determine the co-ordinates of the new point P_{xyz} by homogeneous transformation.

25. Determine the homogeneous transformation matrix to represent a rotation of 30^0 about OX axis and a translation of 8 units along the OB axis of the mobile frame.
26. Determine the homogeneous transformation matrix to represent the following sequence of operations.
    a. Rotation of 60^0 about OX axis
    b. Translation of 4 units along OX axis
    c. Translation of -6 units along OC axis
    d. Rotation of 30^0 OB axis

UNIT-3

1. List the types of drive systems used in robots. Explain briefly each of them.
2. Give a brief classification of actuators used in robots
3. List the advantages and disadvantages of hydraulic drives.
4. List the advantages and disadvantages of pneumatic drives.
5. List the advantages and dis-advantages of Electrical drives.
6. Discuss about the features of the various drive systems for an Industrial robot
7. Discriminate the factors which must be considered while choosing the drive system for robots
8. Analyze about the robot drive system selection procedure in detail.
9. What is end effector? Give some examples and applications of Robot End Effector
10. What is meant by gripper? List the types of grippers. Explain any two types
11. Analyze the working of different types of mechanical grippers.
12. Draw the different mechanism used in the mechanical gripper and describe any two mechanism.
13. Explain vacuum grippers, with reference to the principle and applications
14. Describe the magnetic grippers in robot. Give any two limitations of magnetic grippers
15. Describe the factors to be considered while selecting the grippers for robot
16. Infer important factors to be considered in design of grippers
17. Illustrate gripper force analysis.
18. Enumerate the difference between open loop and closed loop control System
19. Discriminate the salient features, capabilities, applications, merits and limitations of non-servo and servo controlled robots.
20. Describe robot playback concept with continuous path control system
21. Explain briefly point to point contact and continuous path control robot.
22. Derive with usual notations the expression for force exerted by the mechanical gripper in robotics.
23. Explain the gripper design consideration in robotics.
24. Differentiate between reactive and deliberative control system

UNIT-4

1. Classify the methods of Robot programming. Mention the advantages and disadvantages of lead through programming.
2. Classify the different types of programming methods and illustrate any one in detail
3. What is teach pendant? Describe the teach pendant for Robot system
4. What are the ways of accomplishing lead through programming? Explain detail manual lead through programming method in robot application
5. Describe the Powered lead through robot programming and list any one benefit of it.
6. Describe the capabilities of and limitations of lead through programming
7. Describe method of Robot programming with textual languages. Explain the First-generation languages.
8. Illustrate different generation languages in robot programming
9. Briefly explain the Future generation languages of robots.
10. With a neat sketch explain the Robot language structure.
11. Explain the Second-generation languages of robotics.
12. Define sensor. Give classification of sensors with their functions.
13. With neat sketch explain the following: i) Potentiometer. (ii) Optical Incremental Encoders.
14. List the components of Optical encoder and with a neat labelled diagram of Incremental encoder mention its salient features.
15. List the components of Optical encoder and with a neat labelled diagram of absolute encoder mention its salient features.
16. Write the neat labelled diagram of Potentiometer and illustrate its working process.
17. Write the neat labelled diagram of Resolver and illustrate its working process.
18. Illustrate machine vision system concept used in industry.
19. What are the three function of vision system Explain and also mention applications.
20. Illustrate velocity sensors with neat diagram.

UNIT-5

1. Differentiate path planning, motion planning and trajectory planning.
2. What are the various forms of input required to be provided as constraints to the robot trajectory planning? explain
3. Describe different path control modes in robotics.
4. What are the General consideration in trajectory planning? Explain
5. Discuss different Schemes of path generation with advantages and disadvantages.
6. What are the General consideration in Joint space scheme? Explain
7. Enumerate trajectory generation polynomial types.
8. Explain the parameters involved in the path planning with 3rd degree polynomial.
9. Evaluate the advantages of robotic welding and assembly systems in manufacturing industries, considering factors such as precision, speed, and quality control.

10. Explore the applications of robotics in agriculture, such as precision farming, crop monitoring, and harvesting, highlighting their potential to increase yields and reduce resource consumption.
11. Evaluate the role of collaborative robots (cobots) in computer integrated manufacturing, including their impact on worker safety, productivity, and job satisfaction.
12. Analyze the potential impact of autonomous robotics on society, considering factors such as job displacement, safety, and ethical considerations
13. Evaluate the role of Robots in Medical care and Hospitals.
14. Explore the applications of robotics in Industry for material handling and assembly work.
15. Explore the applications of robotics in construction trades
16. Explore the applications of robotics in space explorations
17. Explore the applications of robotics in defense, surveillance and Firefighting